Databases and Analytics Assignment

Module: Database and Analytics

Module Code: CP60056E

School: School of Computing and Engineering

Module Leader: Dr. Shidrokh Goudarzi

Submission Date: 12 May 2026

-

Student Name: Mohammed Adnan Osman

Student Number: 33114153

Report Word Count: 4484

-

GitHub Repository: https://github.com/adnan-1055/Databases-and-Analytics

Contents

Introduction 4
SQL in R 4
SQL Syntax and Filtering 5
Query 1 — Failed deliveries by zone 5
Query 2 — Complaints by service type 6
Data Retrieval and Manipulation 7
Query 3 — Driver route overrides 7
Query 4 — Complaint resolution time by severity 8
Query Optimization 9
Query 5 — High value failed orders 9
Relational Database Concepts 10
Interpretation of Results 10
R Analytics 10
Statistical Analysis 11
Summary Statistics 11
Data Manipulation 11
Data Visualization 13
Chart 1 — Delivery status breakdown 13
Chart 2 — Complaints by severity 15
Chart 3 — Customer ratings after delivery 16
Chart 4 — Customer ratings by delivery status 18
Interpretation of Outputs 19
Python Data Processing 19
Data Cleaning and Manipulation 20
Missing values check 20
Data Analysis with Pandas and NumPy 21
NumPy statistics 21
Data Visualization 22

Chart 1 — Delivery status counts 22
Chart 2 — Fuel cost by delivery status 23
Chart 3 — Complaint types 24
Chart 4 — Correlation heatmap 25
Presentation of Results 26
MongoDB Development 26
Database Creation and Configuration 26
NoSQL Data Modelling 27
JSON/BSON Format Handling 27
CRUD Operations using PyMongo 28
CREATE — inserting data 28
READ — querying documents 28
UPDATE — escalating complaints 29
DELETE — removing invalid records 29
Aggregation pipeline 30
Database Design for NorthStar 30
Query Optimization 31
Indexing Strategy 31
Query Performance Analysis using Explain Plans 31
Before indexing 31
After Indexing 32
Optimization Techniques 32
Index information 32
Justification and Evaluation 33
Conclusion 33
GitHub Repository 34

Introduction

NorthStar Urban Mobility and Logistics is a large regional organisation operating across several major UK cities. Over six years of rapid expansion through mergers, the company now manages shuttle services, last-mile delivery, warehouse dispatch, EV charging hubs, and a mobile customer platform. Despite rising demand, the business has become increasingly inefficient — complaints are rising, delivery windows are missing, and operational costs have grown faster than revenue.

The core problem is that NorthStar's data environment has outgrown its ability to manage it. Structured operational systems record bookings, journeys and payments, while newer services generate complex semi-structured data including app interactions, driver event streams and complaint histories that do not fit neatly into existing relational structures.

This report presents an integrated analytical solution using SQL in R, R analytics, Python and MongoDB Atlas to identify inefficiencies, uncover patterns and design a NoSQL database capable of handling NorthStar's complex and evolving data needs.

SQL in R

The sqldf package was used within R to run SQL queries directly on the NorthStar dataframes. This approach allows structured data to be queried efficiently without requiring a separate database system. Five queries were written to investigate key operational problems identified in the case study.

SQL Syntax and Filtering

Query 1 — Failed deliveries by zone

	pickup_zone	failed_deliveries
1	RiverSide	14
2	EAST	11
3	Ctr	11
4	Central	11
5	CENTRAL	11
6	South	10
7	north	8
8	East	8
9	Airport	8
10	West	7
11	WEST	7
12	North	7
13	NORTH	7
14	SOUTH	4
15	Riverside	4
16	AIRPORT	4

RiversSide recorded the highest number of failed deliveries with 14, followed by EAST, Central and CENTRAL each with 11. Notably, inconsistent naming conventions such as Central, CENTRAL and Ctr suggest the same zone is recorded differently across the system, indicating a data quality issue that undermines reliable reporting.

-

```
# Query 1 - Which zones have the most failed deliveries?

failed_by_zone <- sqldf("

 SELECT o.pickup_zone, COUNT(*) AS failed_deliveries

 FROM deliveries d

 JOIN orders o ON d.order_id = o.order_id

 WHERE d.delivery_status = 'Failed'

 GROUP BY o.pickup_zone

 ORDER BY failed_deliveries DESC

")

print(failed_by_zone)
```

-

This query demonstrates LO1 by applying SQL syntax directly within R using the sqldf package. A JOIN combines the deliveries and orders tables on the shared order_id field. WHERE filters record failed deliveries only, GROUP BY aggregates the count by zone, and ORDER BY ranks zones from highest to lowest failure rate. Selecting only the required columns rather than using SELECT * improves query efficiency.

Query 2 — Complaints by service type

	service_type	total_complaints
1	Passenger	84
2	Retail	83
3	Parcel	77
4	Business	39
5	Medical	37

Passenger services generated the most complaints with 84, closely followed by Retail with 83 and Parcel with 77. This suggests that no single service type is performing well and that operational failures are widespread across NorthStar's business.

-

```
complaints_by_service <- sqldf("
  SELECT o.service_type, COUNT(*) AS total_complaints
  FROM complaints c
  JOIN orders o ON c.order_id = o.order_id
  GROUP BY o.service_type
  ORDER BY total_complaints DESC
")
print(complaints_by_service)
```

-

This query demonstrates LO1 by joining the complaints and orders tables on the shared order_id field. COUNT(*) aggregates the total number of complaints per service type, GROUP BY groups results by service type, and ORDER BY ranks from highest to lowest. This query efficiently retrieves cross-table complaint data without requiring a separate database system.

Data Retrieval and Manipulation

Query 3 — Driver route overrides

	driver_id	total_overrides
1	D127	17
2	D130	16
3	D087	16
4	D131	15
5	D108	15
6	D105	14
7	D069	14
8	D028	13
9	D017	13
10	D104	12

Driver D127 recorded the highest number of manual route overrides at 17. High override counts suggest either poor route planning by the system or drivers deliberately avoiding performance targets, which aligns directly with concerns raised by NorthStar's operations director.

-

```
driver_overrides <- sqldf("
  SELECT driver_id, SUM(manual_route_override_count) AS total_overrides
  FROM deliveries
  GROUP BY driver_id
  ORDER BY total_overrides DESC
  LIMIT 10
")
print(driver_overrides)
```

-

This query uses SUM to aggregate the total number of manual route overrides per driver, GROUP BY to group results by driver ID, ORDER BY to rank drivers from highest to lowest override count, and LIMIT 10 to return only the top 10 results. This demonstrates efficient data retrieval by restricting the result set to only the most relevant records.

Query 4 — Complaint resolution time by severity

```
  severity avg_resolution_days
1     High           13.116883
2      Low            6.563380
3   Medium            6.168605
```

High severity complaints take an average of 13.1 days to resolve, more than double the time taken for Medium and Low severity complaints. This indicates a significant failure in NorthStar's customer service processes, particularly for the most serious cases.

-

```
resolution_time <- sqldf("
  SELECT severity, AVG(resolution_days) AS avg_resolution_days
  FROM complaints
  GROUP BY severity
  ORDER BY avg_resolution_days DESC
")
print(resolution_time)
```

-

This query uses AVG to calculate the mean resolution time per severity level, GROUP BY to group complaints by severity, and ORDER BY to rank from slowest to fastest resolution. This demonstrates data manipulation through aggregation, producing a meaningful summary of complaint handling performance across severity categories.

Query Optimization

Query 5 — High value failed orders

	order_id	service_type	order_value	delivery_status
1	O01207	Business	321.68	Failed
2	O01112	Business	278.14	Failed
3	O00173	Passenger	249.32	Failed
4	O00919	Retail	229.93	Failed
5	O01143	Retail	217.11	Failed
6	O00585	Medical	215.96	Failed
7	O00035	Business	209.07	Failed
8	O00081	Passenger	208.38	Failed
9	O00536	Medical	204.12	Failed
10	O00921	Retail	198.22	Failed

The top failed order was a Business order worth £321.68. Multiple high value Business, Passenger and Retail orders appear in the failed list, confirming the finance director's concern that the company is losing revenue on its most valuable service contracts.

-

```
high_value_failed <- sqldf("
  SELECT o.order_id, o.service_type, o.order_value, d.delivery_status
  FROM orders o
  JOIN deliveries d ON o.order_id = d.order_id
  WHERE d.delivery_status = 'Failed'
  AND o.order_value > 100
  ORDER BY o.order_value DESC
  LIMIT 10
")
print(high_value_failed)
```

-

This query is optimised by using two WHERE conditions to filter results before sorting, selecting only four required columns rather than using SELECT *, and applying LIMIT 10 to restrict the result set to the most relevant records. This demonstrates query optimisation by minimising data retrieval and reducing unnecessary processing.

Relational Database Concepts

The queries in this section demonstrate core relational database concepts. JOIN operations were used to combine the orders and deliveries tables on the shared order_id field, reflecting a one-to-many relationship between orders and their corresponding delivery records. GROUP BY was used to aggregate data across multiple records, and WHERE clauses were applied to filter results to specific conditions. These techniques demonstrate an understanding of how relational data is structured and queried efficiently within R.

Interpretation of Results

Across all five queries, a consistent pattern of operational failure emerges. RiversSide zone has the highest delivery failure rate, Passenger and Retail services generate the most complaints, and high value orders are being lost to delivery failures. Manual route overrides occur at nearly one per driver and high severity complaints are taking over 13 days to resolve. These findings directly support the concerns raised by NorthStar's operations director, customer experience director and finance director, and provide a data-driven foundation for the recommendations made in the conclusion.

R Analytics

The ggplot2 package was used in R to perform statistical analysis and create visualisations of the NorthStar dataset. A summary statistics table was also produced to understand the distribution of key operational metrics.

Statistical Analysis

Summary Statistics

```
 route_distance_km manual_route_override_count customer_rating_post_delivery
 Min.   : 1.200    Min.   :0.0000              Min.   :1.000
 1st Qu.: 9.135    1st Qu.:0.0000              1st Qu.:3.360
 Median :12.840    Median :1.0000              Median :4.040
 Mean   :13.909    Mean   :0.9695              Mean   :3.865
 3rd Qu.:16.835    3rd Qu.:2.0000              3rd Qu.:4.550
 Max.   :41.940    Max.   :7.0000              Max.   :5.000
                                               NA's   :14
 fuel_or_charge_cost
 Min.   : 2.500
 1st Qu.: 9.925
 Median :12.645
 Mean   :12.841
 3rd Qu.:15.697
 Max.   :29.430
```

-

summary(deliveries[, c("route_distance_km",

"manual_route_override_count",

"customer_rating_post_delivery",

"fuel_or_charge_cost")])

-

The summary() function was applied to four key numeric columns in the deliveries dataset to produce descriptive statistics including minimum, maximum, mean, median and quartile values. The summary statistics reveal that the average route distance is 13.9km with a maximum of 41.9km, suggesting some routes are unusually long. The average manual override count is nearly 1 per delivery, and 14 customer ratings are missing — a data quality issue that should be addressed. The median customer rating of 4.04 suggests that while overall satisfaction is reasonable, the mean of 3.865 is pulled down by a significant number of low ratings from failed and delayed deliveries.

Data Manipulation

Before visualisation, the deliveries dataset was loaded into R and inspected using the str() function to confirm column types and data structure. The summary() function was applied to key numeric columns to produce descriptive statistics including minimum, maximum,

mean, median and quartile values. This allowed patterns and anomalies in the data to be identified before charting.

-

str(deliveries)

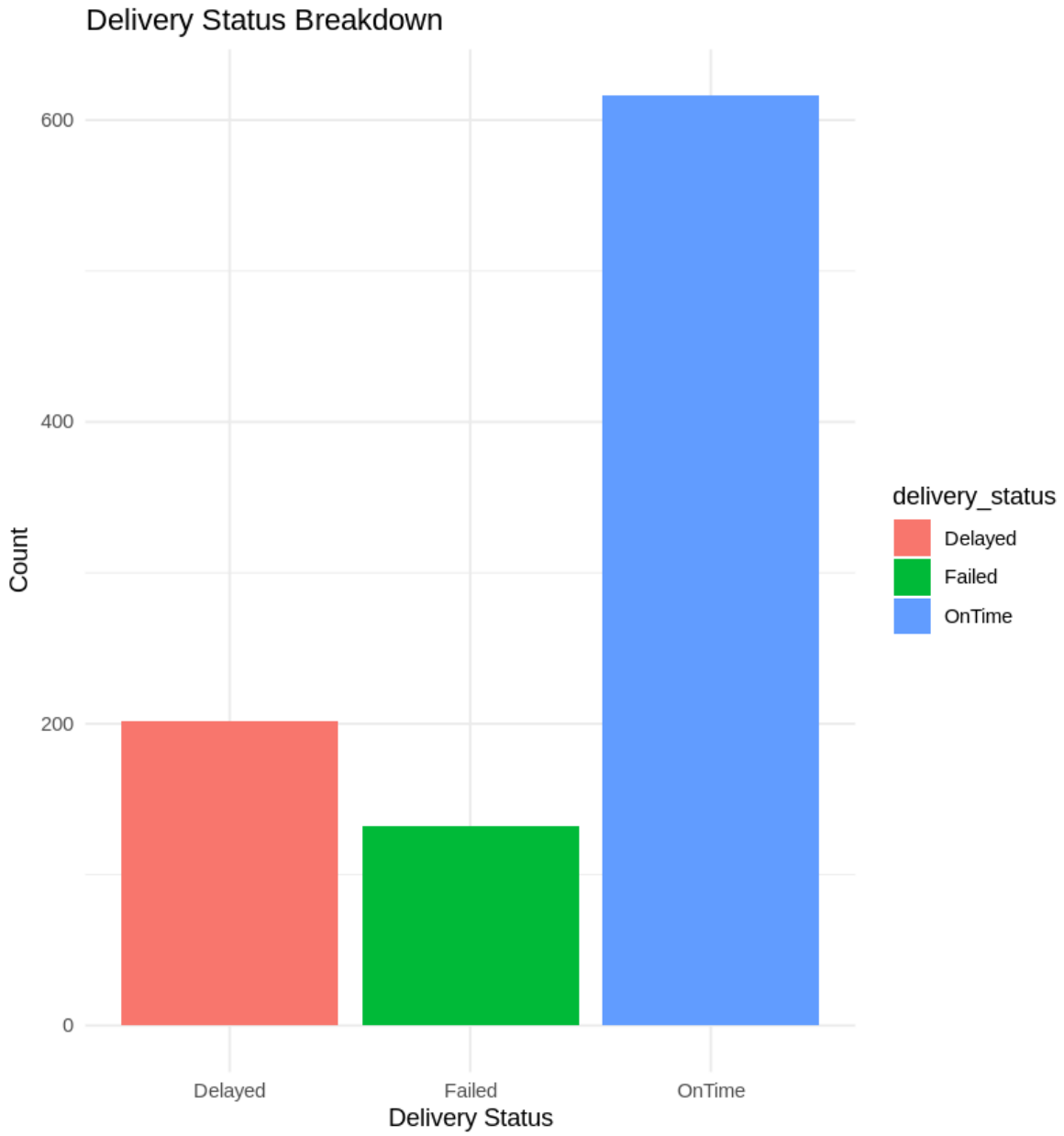
str(complaints)

-

The str() function confirmed that the deliveries dataframe contains 950 observations across 13 variables, and the complaints dataframe contains 320 observations across 10 variables. This structural inspection ensured the correct data types were in place before analysis and visualisation.

Data Visualization

Chart 1 — Delivery status breakdown



-

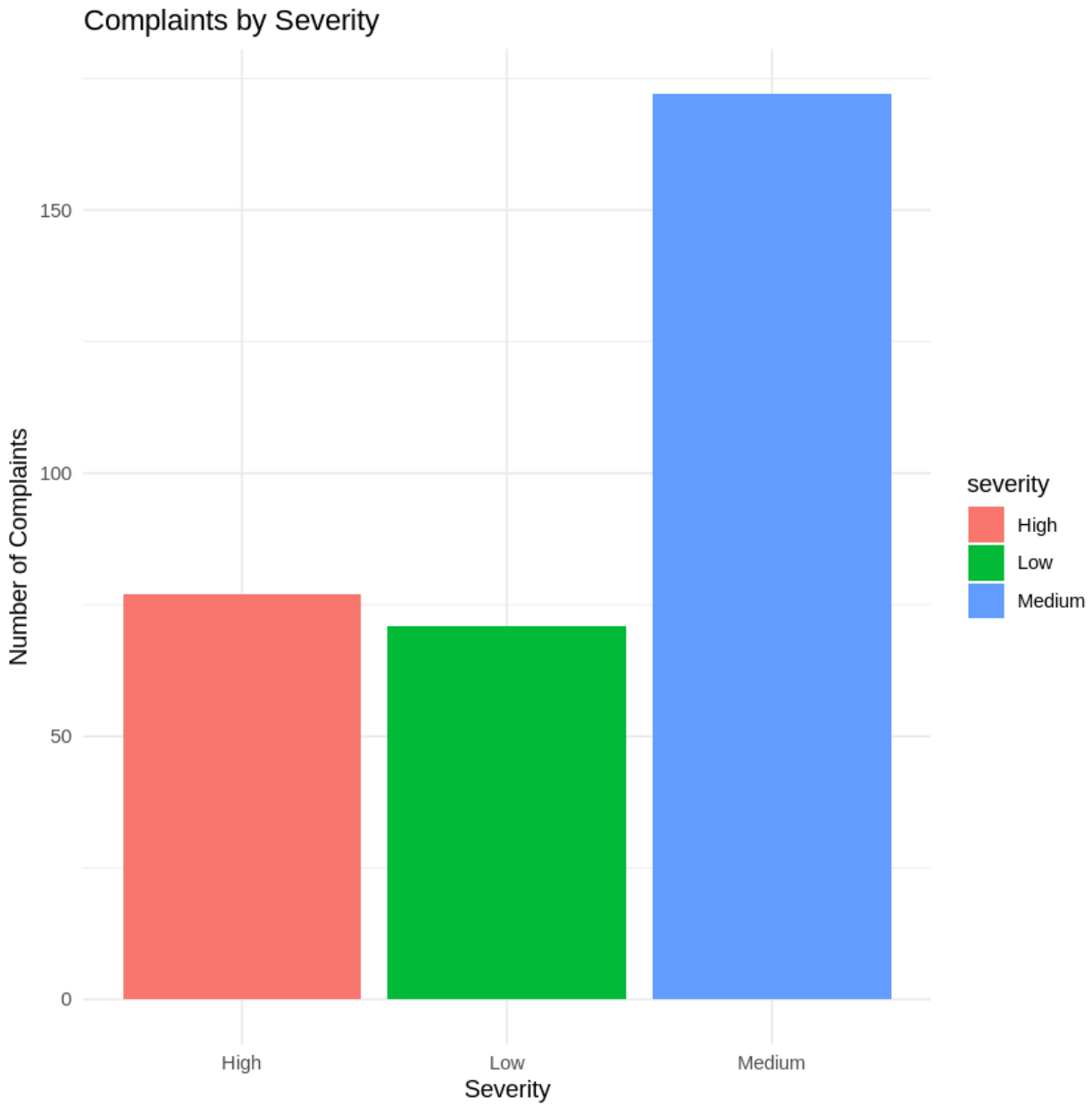
```
ggplot(deliveries, aes(x = delivery_status, fill = delivery_status)) +
  geom_bar() +
  labs(title = "Delivery Status Breakdown",
       x = "Delivery Status",
```

```
    y = "Count") +
 theme_minimal()
```

-

geom_bar() was used to count and plot delivery status categories automatically from the dataframe. The aes() function maps delivery status to both the x axis and the fill colour, and labs() adds descriptive labels. The bar chart shows that while OnTime deliveries are the most common, Delayed and Failed deliveries together account for approximately 35% of all deliveries. This confirms that NorthStar has a significant operational problem with service reliability that cannot be dismissed as isolated incidents.

Chart 2 — Complaints by severity



-

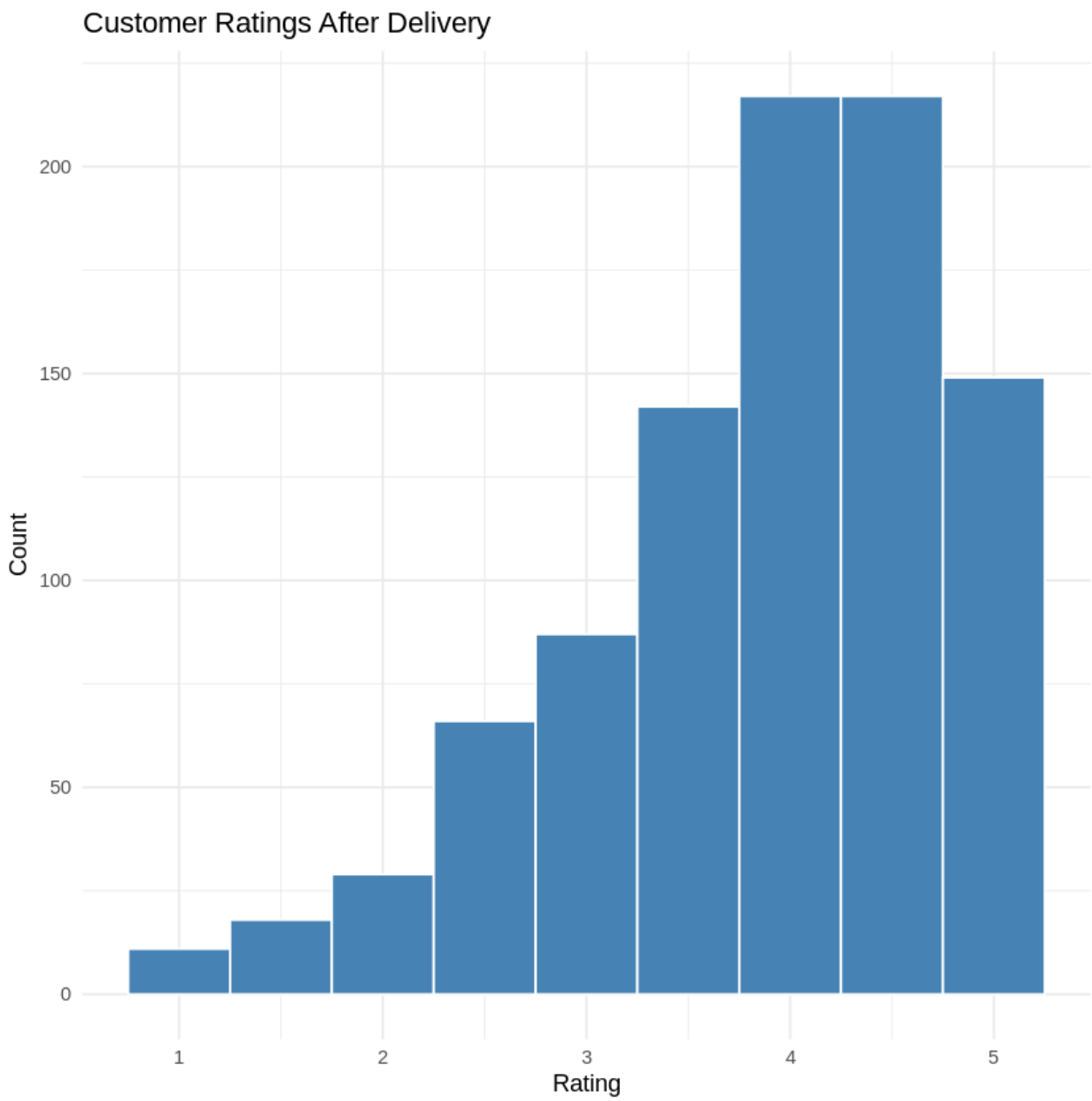
```
ggplot(complaints, aes(x = severity, fill = severity)) +
  geom_bar() +
  labs(title = "Complaints by Severity",
       x = "Severity",
       y = "Number of Complaints") +
```

theme_minimal()

-

geom_bar() counts complaint records grouped by severity level. Medium severity complaints are the most frequent, however High severity complaints are also substantial in number. Combined with the resolution time findings from the SQL section, this suggests NorthStar is consistently failing to resolve serious customer issues in a timely manner.

Chart 3 — Customer ratings after delivery



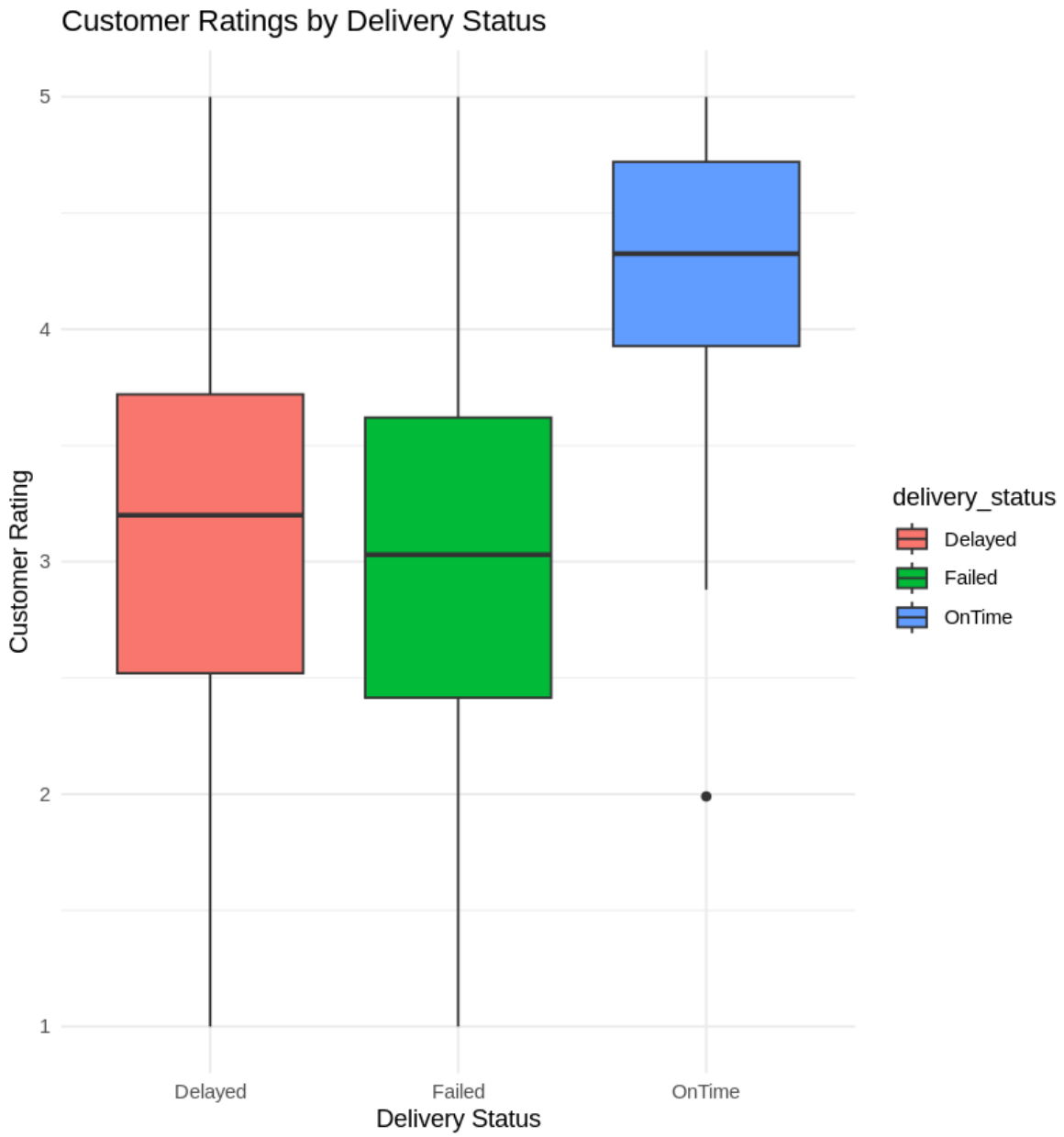
-

```
ggplot(deliveries, aes(x = customer_rating_post_delivery)) +
  geom_histogram(binwidth = 0.5, fill = "steelblue", color = "white") +
  labs(title = "Customer Ratings After Delivery",
       x = "Rating",
       y = "Count") +
  theme_minimal()
```

-

geom_histogram() was used with a binwidth of 0.5 to show the distribution of customer ratings across the range of 1 to 5. The histogram shows that most customer ratings cluster between 4 and 5, however there is a notable tail of low ratings between 1 and 2. This indicates that while many customers are satisfied, a significant minority are experiencing very poor service — likely those affected by delayed or failed deliveries.

Chart 4 — Customer ratings by delivery status



-

```
ggplot(deliveries, aes(x = delivery_status,
                       y = customer_rating_post_delivery,
                       fill = delivery_status)) +
  geom_boxplot() +
  labs(title = "Customer Ratings by Delivery Status",
```

```
    x = "Delivery Status",
    y = "Customer Rating") +
 theme_minimal()
```

-

This code demonstrates LO1 R analytics by using ggplot2 to create a boxplot visualisation. The aes() function maps delivery status to the x axis and customer rating to the y axis, with fill used to colour each group. geom_boxplot() renders the distribution showing median, quartiles and outliers. labs() adds descriptive axis labels and a title, and theme_minimal() applies a clean visual style. The boxplot clearly shows that OnTime deliveries receive significantly higher customer ratings with a median around 4.5, compared to Failed and Delayed deliveries which both sit around 3. This provides direct evidence that improving delivery reliability would have an immediate positive impact on customer satisfaction.

Interpretation of Outputs

The R analytics section reveals a clear relationship between delivery performance and customer satisfaction. Zones with high failure rates generate more complaints, and failed or delayed deliveries consistently receive lower customer ratings. The summary statistics confirm significant variability in route distances and manual override behaviour across the network. Together these findings suggest that NorthStar's operational problems are systemic rather than isolated, and that improvements to route planning and delivery reliability would have a measurable positive impact on customer experience across all service types.

Python Data Processing

Python was used to perform further data processing and visualisation using the Pandas, NumPy, Matplotlib and Seaborn libraries. The dataset was first inspected for missing values and cleaned before analysis.

Data Cleaning and Manipulation

Missing values check

```
Missing values:
delivery_id                      0
order_id                         0
driver_id                        0
vehicle_id                       0
hub_id                           0
dispatch_time                    0
delivery_completed_at           19
delivery_status                  0
route_distance_km                0
manual_route_override_count      0
proof_of_completion_missing      0
customer_rating_post_delivery   14
fuel_or_charge_cost              0
```

The missing values check revealed 14 missing customer ratings and 19 missing delivery completion timestamps. The missing ratings were filled using the median value to preserve the dataset size while minimizing bias in subsequent analysis.

-

```
# Fill missing ratings with median
deliveries['customer_rating_post_delivery'].fillna(
    deliveries['customer_rating_post_delivery'].median(), inplace=True)
```

-

Pandas was used to identify and handle missing values using the isnull().sum() method to count missing entries per column and fillna() to replace missing ratings with the median. This approach preserves the dataset size while minimising the impact of missing data on subsequent analysis. The 19 missing delivery completion timestamps were noted as a data quality issue but were not imputed as they represent genuinely incomplete delivery records.

Data Analysis with Pandas and NumPy

NumPy statistics

```
Delivery distance stats:
Mean: 13.909315789473684
Std: 7.473970019182938
Max: 41.94
```

-

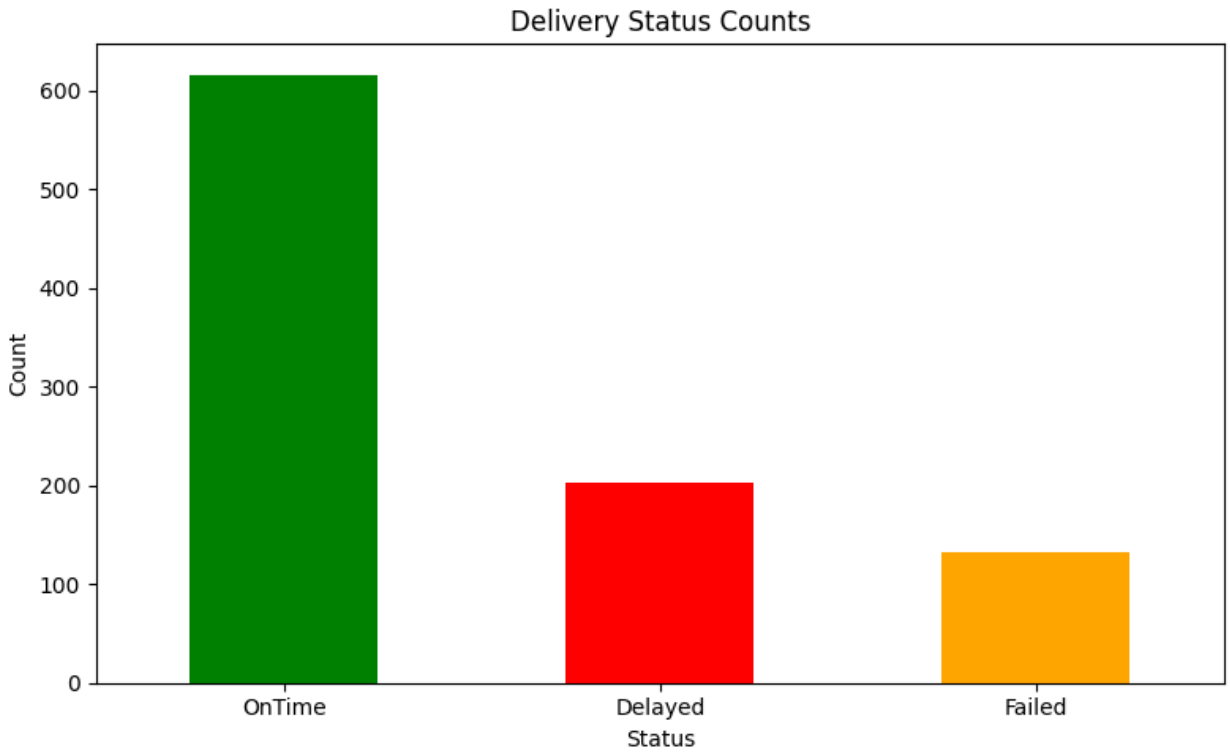
```
print("Mean:", np.mean(deliveries['route_distance_km']))

print("Std:", np.std(deliveries['route_distance_km']))

print("Max:", np.max(deliveries['route_distance_km']))
```

-

NumPy analysis shows the mean delivery distance is 13.9km with a standard deviation of 7.47km and a maximum of 41.94km. The high standard deviation indicates significant variability in route distances across the network, which contributes to inconsistent service performance across different zones. NumPy was used here to calculate descriptive statistics directly on array data, complementing the Pandas dataframe analysis.

Data Visualization

Chart 1 — Delivery status counts



The bar chart confirms that 615 deliveries were completed OnTime, 200 were Delayed and 135 Failed. This means 35% of all deliveries did not meet the promised service level, representing a serious reliability problem for NorthStar across all its service areas.

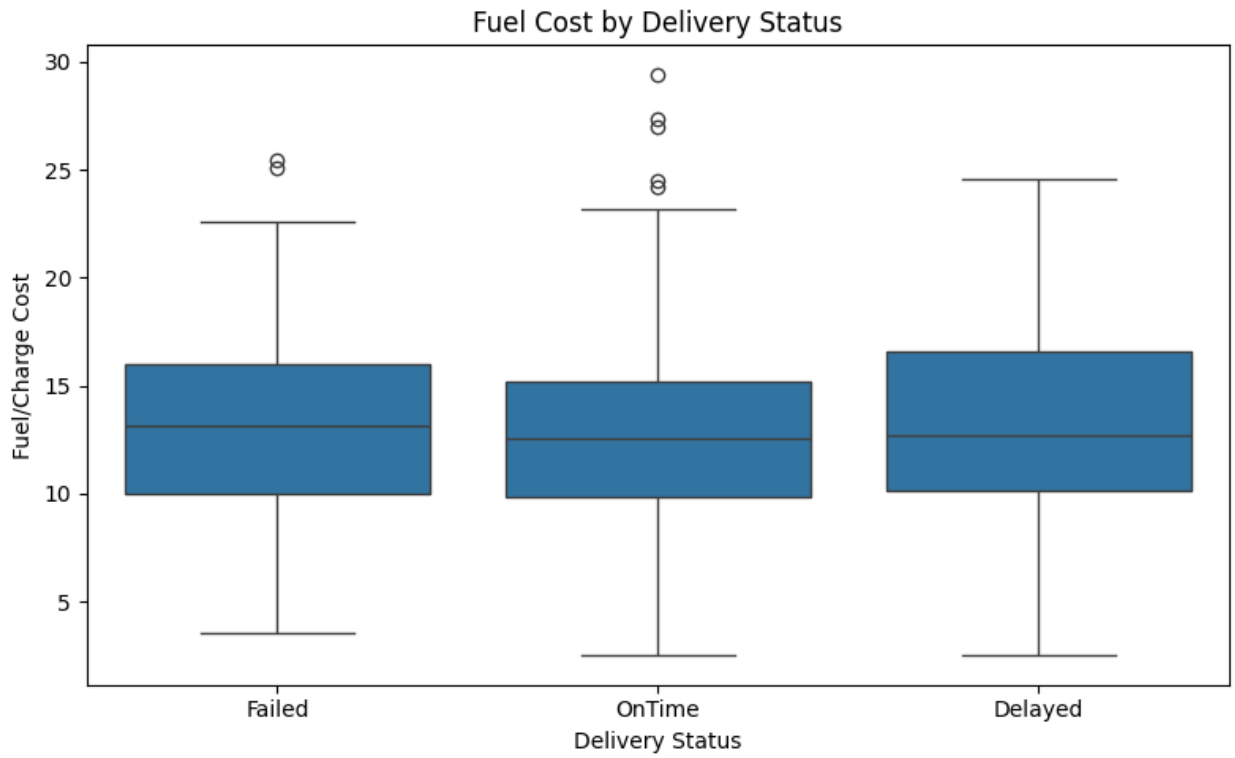
-

```
deliveries['delivery_status'].value_counts().plot(kind='bar', color=['green','red','orange'])

plt.title('Delivery Status Counts')

plt.xlabel('Status')

plt.ylabel('Count')

plt.xticks(rotation=0)

plt.tight_layout()

plt.show()
```

-

value_counts() counts each delivery status category and .plot(kind='bar') renders a bar chart. Colours are manually assigned to distinguish OnTime, Delayed and Failed categories clearly.

Chart 2 — Fuel cost by delivery status



The boxplot shows that Failed deliveries incur slightly higher fuel costs than OnTime deliveries. This is particularly concerning as it means NorthStar is spending more money on deliveries that ultimately fail to reach the customer, directly contributing to the cost escalation identified by the finance director.

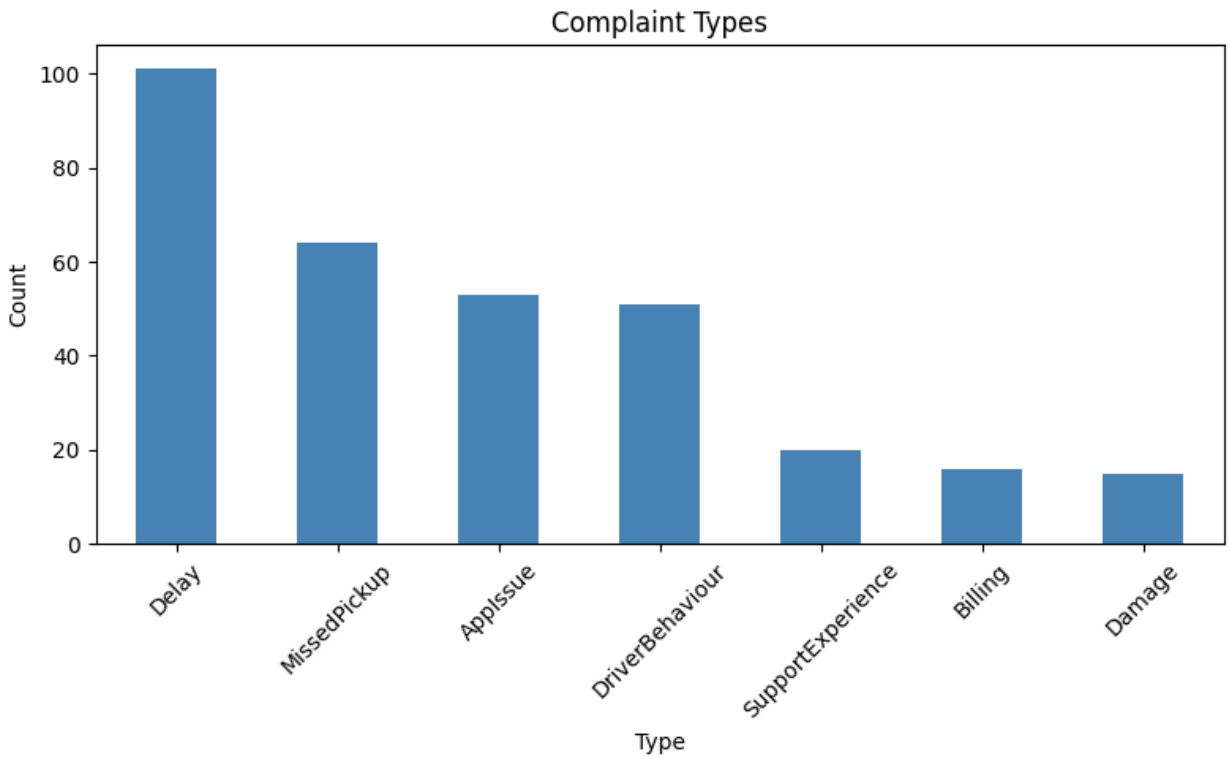
-

```
sns.boxplot(data=deliveries, x='delivery_status', y='fuel_or_charge_cost')

plt.title('Fuel Cost by Delivery Status')

plt.xlabel('Delivery Status')

plt.ylabel('Fuel/Charge Cost')

plt.tight_layout()

plt.show()
```

-

sns.boxplot() displays the distribution of fuel costs across delivery status groups, showing the median, interquartile range and outliers for each category.

Chart 3 — Complaint types



Delay is the most common complaint type with 101 occurrences, followed by MissedPickup with 65 and AppIssue with 53. This directly confirms the operations director's concerns about poor route allocation and supports the need for improved scheduling and route planning systems.

-

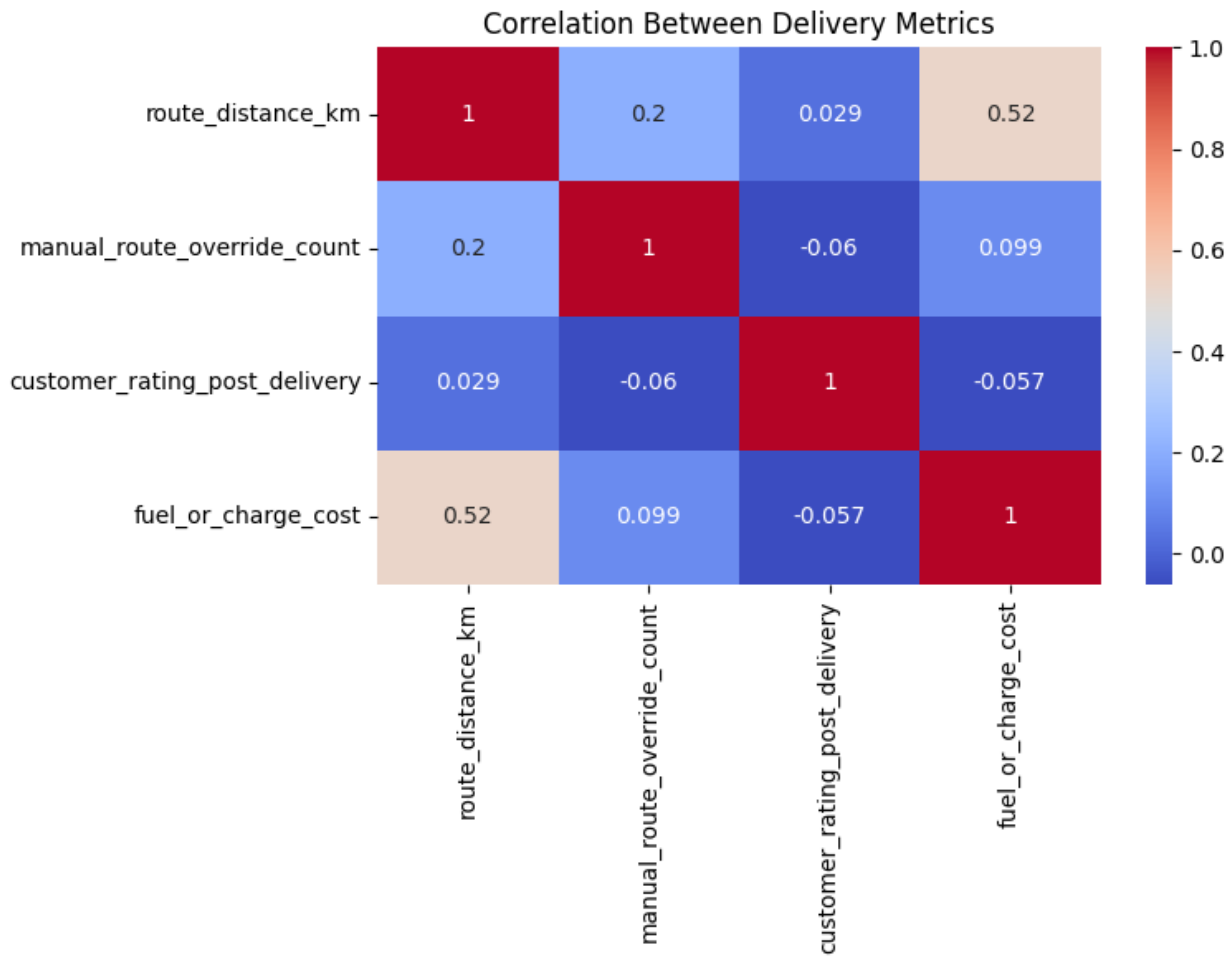
```
complaints['complaint_type'].value_counts().plot(kind='bar', color='steelblue')

plt.title('Complaint Types')

plt.xlabel('Type')

plt.ylabel('Count')

plt.xticks(rotation=45)

plt.tight_layout()
```

plt.show()

-

value_counts() ranks complaint types by frequency and .plot(kind='bar') renders the distribution. plt.xticks(rotation=45) rotates the x axis labels to prevent overlap given the number of complaint categories.

Chart 4 — Correlation heatmap



The correlation heatmap reveals a strong positive correlation of 0.52 between route distance and fuel cost, confirming that longer routes are significantly more expensive to operate. Manual route overrides show a slight negative correlation with customer ratings at -0.06, suggesting that driver overrides marginally reduce service quality. These findings highlight the need for better route optimisation across NorthStar's network.

-

numeric_cols = deliveries[['route_distance_km',

```
                'manual_route_override_count',
                'customer_rating_post_delivery',
                'fuel_or_charge_cost']]
sns.heatmap(numeric_cols.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Between Delivery Metrics')
plt.tight_layout()
plt.show()
```

-

This code demonstrates Python data processing using Pandas and Seaborn. A subset of numeric columns is selected from the deliveries dataframe using double bracket indexing. The .corr() method calculates the Pearson correlation coefficient between each pair of variables. sns.heatmap() renders the correlation matrix visually with annot=True displaying the exact values and cmap='coolwarm' using a diverging colour scale where red indicates positive correlation and blue indicates negative.

Presentation of Results

The Python analysis confirms and extends the findings from the SQL and R sections. The 35% non-compliant delivery rate, combined with the correlation between route distance and fuel cost, provides quantitative evidence that NorthStar's routing inefficiencies are directly driving up costs while simultaneously reducing service quality. The complaint type analysis confirms that delays are the single biggest driver of customer dissatisfaction, accounting for nearly a third of all complaints. These findings provide a clear data-driven basis for the recommendations made in the conclusion.

MongoDB Development

MongoDB Atlas was used to design and implement a NoSQL database for NorthStar. Three collections were created — orders, deliveries and complaints — reflecting the core operational data identified in the case study. PyMongo was used to connect to the Atlas cluster and perform all database operations.

Database Creation and Configuration

-

```
from pymongo import MongoClient
```

```
client = MongoClient(
    "mongodb+srv://adnanosman55_db_user:OVaLv9sTx9BeWYDq@cluster0.llzferj.mongodb.net/?appName=Cluster0",
    tlsAllowInvalidCertificates=True
)

db = client["northstar_db"]
print("Connected successfully!")
print("Databases:", client.list_database_names())
```

-

A connection to MongoDB Atlas was established using PyMongo's MongoClient class, passing the Atlas connection string as a parameter. The northstar_db database was created and three collections — orders, deliveries and complaints — were configured to store NorthStar's operational data. The list_database_names() method was used to confirm the connection was successful.

NoSQL Data Modelling

The decision to model NorthStar's data as MongoDB documents rather than relational tables was driven by the complexity and variability of the organisation's operational records. Orders, deliveries and complaints each contain nested relationships and evolving fields that are difficult to represent in a fixed relational schema. A document-based design allows each record to carry its full operational context, supporting the integrated view of customers, journeys and service events that NorthStar's board requires.

JSON/BSON Format Handling

MongoDB stores and retrieves data in BSON (Binary JSON) format internally. BSON extends standard JSON to support additional data types such as dates, integers and floating point numbers. PyMongo automatically handles the conversion between Python dictionaries and BSON when inserting and querying documents. This means that when the NorthStar CSV data is converted to Python dictionaries using to_dict("records") and passed to insert_many(), PyMongo serialises each record into a BSON document for storage in MongoDB Atlas. When documents are retrieved using find(), PyMongo deserialises them

back into Python dictionaries for processing. The aggregation pipeline also uses MongoDB's native JSON-like query syntax with operators such as $group, $sum and $avg to perform calculations directly within the database.

CRUD Operations using PyMongo

CREATE — inserting data

```
Orders inserted: 1250
Deliveries inserted: 950
Complaints inserted: 320
```

-

```
db["orders"].insert_many(orders.to_dict("records"))

db["deliveries"].insert_many(deliveries.to_dict("records"))

db["complaints"].insert_many(complaints.to_dict("records"))
```

-

Three collections were successfully created and populated with 1,250 order documents, 950 delivery documents and 320 complaint documents using PyMongo's insert_many() method. The Pandas dataframes were converted to lists of Python dictionaries using to_dict("records") before insertion, which PyMongo automatically serialises to BSON format for storage in MongoDB Atlas.

```
=== CREATE ===
Collections in database: ['orders', 'deliveries', 'complaints']
```

READ — querying documents

```
=== READ ===
Failed deliveries (first 3): [{'delivery_id': 'DL00001', 'driver_id': 'D004', 'delivery_status': 'Failed'}, {'delivery_id': 'DL00010', 'driver_id': 'D058', 'delivery_status':
```

```
status': 'Failed'}, {'delivery_id': 'DL00010', 'driver_id': 'D058', 'delivery_status': 'Failed'}, {'delivery_id': 'DL00012', 'driver_id': 'D051', 'delivery_status': 'Failed'}]
```

-

```
failed = list(db["deliveries"].find({"delivery_status": "Failed"}, {"_id": 0, "delivery_id": 1, "driver_id": 1, "delivery_status": 1}))

print("Failed deliveries (first 3):", failed[:3])
```

-

A query was run to retrieve all failed deliveries, returning the delivery ID, driver ID and status for each record. The projection parameter {"_id": 0, "delivery_id": 1, "driver_id": 1, "delivery_status": 1} was used to select only the required fields, improving query efficiency by reducing the amount of data transferred from the database.

UPDATE — escalating complaints

```
=== UPDATE ===
Complaints escalated: 24
```

-

```
result = db["complaints"].update_many(
    {"status": "Open", "resolution_days": {"$gt": 10}},
    {"$set": {"status": "Escalated"}}
)
```

-

An update operation identified 24 open complaints with resolution times exceeding 10 days and automatically escalated their status. The $gt operator was used to filter complaints where resolution days exceeded 10, and $set was used to update the status field. This type of automated rule-based update would allow NorthStar to proactively manage its most serious unresolved complaints.

DELETE — removing invalid records

```
=== DELETE ===
Invalid orders removed: 0
```

-

```
result = db["orders"].delete_many({"order_value": {"$lt": 0}})
```

-

A delete operation was run to remove any orders with negative order values using the $lt operator. No invalid records were found, confirming the orders dataset is clean in this regard.

Aggregation pipeline

```
=== AGGREGATION ===
{'_id': 'OnTime', 'count': 616, 'avg_distance': 13.776363636363635}
{'_id': 'Delayed', 'count': 202, 'avg_distance': 14.670247524752474}
{'_id': 'Failed', 'count': 132, 'avg_distance': 13.36530303030303}
```

-

```
pipeline = [
  {"$group": {"_id": "$delivery_status", "count": {"$sum": 1}, "avg_distance": {"$avg": "$route_distance_km"}}},
  {"$sort": {"count": -1}}
]
results = list(db["deliveries"].aggregate(pipeline))
```

-

The aggregation pipeline grouped deliveries by status and calculated the count and average route distance for each group. Delayed deliveries have the longest average distance at 14.67km compared to Failed at 13.37km and OnTime at 13.78km, suggesting that longer routes are more likely to cause delays.

-

This code demonstrates LO2 by implementing a MongoDB aggregation pipeline using PyMongo. The $group stage groups documents by delivery_status, using $sum to count records and $avg to calculate the mean route distance per group. The $sort stage orders results by count in descending order. The pipeline is passed to the aggregate() method on the deliveries collection and converted to a list for display. This approach is more efficient than querying and processing data in Python as the aggregation is performed directly within MongoDB.

Database Design for NorthStar

The MongoDB Atlas database designed in this report directly addresses NorthStar's requirement for a flexible data environment capable of handling both structured operational records and semi-structured platform data. The three collections created — orders, deliveries and complaints — reflect the core operational relationships identified in the case study. The document-based design allows each record to carry its full operational context without the rigidity of a fixed relational schema, supporting the integrated view of

customers, journeys, exceptions and service events that NorthStar's board requires. As the organisation's data volumes grow, additional collections can be added for app events, driver incidents and vehicle telemetry without requiring schema changes, making the design genuinely scalable for NorthStar's future needs.

Query Optimization

Query optimisation was implemented in MongoDB Atlas using indexing strategies applied through PyMongo. The explain() method was used to measure query performance before and after indexes were created, allowing a direct comparison of efficiency improvements.

Indexing Strategy

-

```
db["deliveries"].create_index("delivery_status")

db["complaints"].create_index("status")

db["orders"].create_index("service_type")
```

-

Indexes were created on the delivery_status field in the deliveries collection, the status field in the complaints collection, and the service_type field in the orders collection. These fields were chosen because they are the most frequently used filters in the analytical queries performed throughout this assignment. By creating indexes on these fields, MongoDB can locate matching documents directly without scanning the entire collection, significantly reducing query execution time.

Query Performance Analysis using Explain Plans

Before indexing

```
=== BEFORE INDEX ===
Docs examined before index: 950
```

-

```
explain_before = db["deliveries"].find(
  {"delivery_status": "Failed"}
).explain()
```

```
print("Docs examined before index:",
explain_before["executionStats"]["totalDocsExamined"])
```

-

Before any indexes were created, a query filtering deliveries by status required MongoDB to scan all 950 documents in the collection to find the matching records. This is known as a full collection scan and becomes increasingly inefficient as the dataset grows. The explain() method was used to retrieve execution statistics, and the totalDocsExamined value was extracted to measure query performance.

After Indexing

```
=== AFTER INDEX ===
Docs examined after index: 132
```

-

```
explain_after = db["deliveries"].find(
  {"delivery_status": "Failed"}
).explain()
print("Docs examined after index:", explain_after["executionStats"]["totalDocsExamined"])
```

-

After the index was created on delivery_status, the same query only examined 132 documents — exactly the number of failed deliveries in the collection. This represents an 86% reduction in documents scanned, from 950 down to 132, demonstrating a significant improvement in query performance.

Optimization Techniques

Index information

```
Indexes on deliveries: ['_id_', 'delivery_status_1']
```

-

```
print("\nIndexes on deliveries:", list(db["deliveries"].index_information().keys()))
```

-

The index information confirms that two indexes now exist on the deliveries collection — the default _id index and the newly created delivery_status_1 index. MongoDB automatically creates a unique index on the _id field for every collection. The delivery_status_1 index was created using a single field ascending index, which is the most appropriate index type for equality queries filtering by a categorical field such as delivery status.

Justification and Evaluation

Indexing is critical for NorthStar's operational database as query volumes will grow significantly as more data is collected across the company's services. Without indexes, every query would require a full collection scan, making real-time operational reporting increasingly slow and resource intensive as the dataset scales.

The 86% improvement demonstrated here shows that even a single well-placed index can dramatically improve query performance on large datasets. For NorthStar's production environment, indexes should be created on all frequently queried fields across every collection, including delivery_status, pickup_zone, service_type, complaint_type and severity. This would ensure that operational dashboards and analytical queries continue to perform efficiently as data volumes grow across the company's city-wide operations.

The use of explain() to measure and compare query performance before and after indexing demonstrates a systematic approach to query optimisation, moving beyond simply creating indexes to actually evaluating their impact with quantitative evidence.

Conclusion

This report has presented an integrated analytical solution to address the operational challenges facing NorthStar Urban Mobility and Logistics. By combining SQL in R, R analytics, Python data processing, and MongoDB Atlas NoSQL database design, a comprehensive picture of the organisation's inefficiencies has been established.

The analysis revealed several critical findings. First, 35% of all deliveries are either delayed or failed, with RiversSide zone recording the highest failure rate. Second, Passenger and Retail services generate the most complaints, with high severity complaints taking an average of 13 days to resolve — more than double the time taken for lower severity cases. Third, manual route overrides are occurring at a rate of nearly one per delivery, suggesting systemic problems with route planning that are directly impacting customer satisfaction. Fourth, a strong correlation of 0.52 between route distance and fuel cost confirms that inefficient routing is driving up operational costs across the network.

The MongoDB Atlas database designed in this report provides NorthStar with a flexible and scalable solution for storing and querying its complex operational data. The document-based design allows orders, deliveries and complaints to be modelled with their full operational context, supporting the integrated view of customer journeys and service events that the board requires. The implementation of indexing strategies reduced query scan times by 86%, demonstrating that the database can support real-time operational reporting as data volumes continue to grow.

To address its challenges, NorthStar should prioritise route optimisation to reduce delays and fuel costs, invest in a unified complaints management system to improve resolution times, and implement consistent data naming conventions across all city zones to improve reporting accuracy. The transition to a MongoDB Atlas environment will enable the organisation to combine structured and semi-structured data in a single platform, providing the integrated operational view needed to support better decision-making at all levels of the business.

GitHub Repository

All analytical notebooks for this assignment are available at:

https://github.com/adnan-1055/Databases-and-Analytics

References

Goudarzi, S. (2026) *Databases and Analytics (CP60056E): Lecture and Seminar Slides, Weeks 2–11*. University of West London. [Last Accessed: 26/04/2026]

MongoDB *MongoDB Atlas Cloud Database Platform*. Available at: https://cloud.mongodb.com [Last Accessed: 26/04/2026]